



Discrete Optimization

Classifying and modeling setups and cleanings in lot sizing and scheduling

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ABSTRACT

Much attention in the lot sizing and scheduling literature has been focused on reducing the number and size of setups. Cleanings, in contrast, remain a key cost driver in large parts of the process industries such as the food and pharmaceutical sectors. Here, quality and safety considerations lead to a diversity of cleaning requirements. A prerequisite for an efficient use of resources is an accurate representation of the constraints imposed by the different setups and cleanings. In this paper, we therefore first develop a general classification scheme for setups and cleanings. Three different classes are identified: batch-, time-, and volume-dependent setups and cleanings. The classes are further differentiated based on their separability, substitutability, reference point, flexibility, product dependency, and batch-size dependency. Secondly, we develop a generic optimization model for lot sizing and scheduling in the typical process-industry setting of flowshops, accurately representing all setup and cleaning classes. Thirdly, we apply the model to the case of cheese production in no-wait flowshops, demonstrating the adaptability of the generic model to industry-specific settings as well as the computational efficiency of the approach. The results show that significant reductions in machine downtime and makespan are achieved. Finally, our numerical tests provide insights into the extent to which a misidentification of cleaning classes may decrease scheduling flexibility and impair solution quality. Our results also show the benefits of considering the heterogeneity in processing times, which can be used to compensate for setup- and cleaning-time differences during successive production stages.

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1. Introduction

In the design and operation of production systems, a significant effort is normally made to reduce the amount of non-value-adding activities. Recent developments for instance include reducing changeover time by conducting setups simultaneously with production activities (Allahverdi & Soroush, 2008). However, in the process industries, changeovers mainly consist of cleanings. They must be performed while production is stopped. Quality and safety regulations, for instance in the pharmaceutical, chemical, and food sectors, also prevent significant reduction. Cleanings cause long downtimes for machines, waste material, and consume energy, water, and cleaning agents. Together, these are key cost drivers in industries often characterized by high volumes and low profit margins. For lot sizing and scheduling in the process industries, it is therefore fundamental to account for cleanings in addition to setups.

There are many different classes of setups and cleanings that require different approaches in lot sizing and scheduling. Among these are setups and cleanings when switching between two product batches and those when a certain production time or volume has been reached. Furthermore, setup and cleaning classes are often interrelated. Cleanings of different intensities may, for instance, be substitutable. A systematic classification of cleanings and their specific characteristics is however lacking. Such a classification is prerequisite for developing a generic lot sizing and scheduling approach for setups and cleanings. Only a correct representation of the constraints that setups and cleanings impose enables an efficient use of resources. An incorrect formulation leads either to infeasibility or suboptimality. In particular, a formulation that is too rigid reduces problem complexity but also over-constrains the solution space. As a result, the lot sizing and scheduling problem solution cannot exploit the actual flexibility of the manufacturing system. In the application of the modeling approach, the correct identification of the setup and cleaning classes is crucial.

In the process industries, the prevalent type of production systems are flowshops. On an aggregate level, process industry production systems are often modeled as two-machine flowshops.

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The first machine may for example reflect continuous processing, whereas the second machine may reflect the transfer of the continuous flow into a flow of discrete units. An example of such flowshops are make-and-pack systems, frequently used in the production of consumer goods. The development of a generic lot sizing and scheduling approach covering setups and cleanings in two-machine flowshops can therefore be applied in many process industry settings. Moreover, it can easily be extended to multiple machines by re-utilizing the modeling concepts that link the production stages.

Extensive cleaning requirements typify the food sector. Important aspects for lot sizing and scheduling in this industry include precedence relationships among products, heterogeneous processing times, as well as a variety of setups and cleanings. The example of cheese production exhibits a combination of the characteristics. Cleanings consume a significant share of the of the available production time (typically more than 20%). Several types of cleanings are required, for example, when switching between products and after a certain production time. As the core processes for cheese production can be modeled as a two-machine flowshop with setups and cleanings, this environment is a suitable illustration for the lot sizing and scheduling approach presented in this paper. However, also problem specifics exist, for which a generic modeling approach must be adapted. In cheese production, a no-wait condition applies between the production stages and the usage of forms, into which the cheeses are filled, is an important consideration.

The main contributions of this paper are (i) the development of a classification scheme for setups and cleanings; (ii) the development of an adaptable Mixed Integer Linear Programming (MILP) model for lot sizing and scheduling of flowshops with comprehensive representation of setups and cleanings, including their interrelationships; (iii) the application of the model to a real case at a medium-sized German cheese dairy with additional no-wait condition and capacity restriction on form usage; and (iv) the analysis of the value of the approach depending on the flexibility in scheduling cleanings and the heterogeneity of the processing times.

The remainder of the paper is organized as follows. [Section 2](#) presents the developed classification scheme for setups and cleanings. Note that the term "changeovers" is synonymous with "setups and cleanings" in the paper. [Section 3](#) outlines the related literature. [Section 4](#) presents the developed modeling approach. The application of the model to cheese production is described in [Section 5](#), including supplementary modeling that captures industry-specific requirements. Furthermore, the developed approach is solved for a case study at a medium-sized German cheese dairy and numerical analyses are carried out. Finally, [Section 6](#) concludes the paper and outlines further research directions.

2. Classification scheme for changeovers

We develop a general classification scheme for changeovers. The classification scheme is inspired by classifications presented in several reviews on scheduling with setup operations. [Allahverdi, Gupta, and Aldowaisan \(1999\)](#), and [Allahverdi, Ng, Cheng, and Kovalyov \(2008\)](#) distinguish setups for batch (family) operations and non-batch (individual jobs) operations, as well as sequence-independent and sequence-dependent setups. [Yang \(1999\)](#) also considers the separability of setup operations. [Cheng, Gupta, and Wang \(2000\)](#) include removal activities after production operations, which are typical in discrete manufacturing flowshops. Finally, [Floudas and Lin \(2004\)](#) focus specifically on chemical processes, for which they also consider time- and frequency-dependent setups. To summarize, some differentiating

setup characteristics have been considered, but none of these reviews has developed a comprehensive classification scheme for changeovers. Amongst other things, the existing schemes cannot be used to capture the scheduling-relevant characteristics of cleanings. Our scheme is a new and unique way of classifying changeovers. The scheme includes characteristics that have not been captured in other schemes, like substitutability, reference point, flexibility, and batch-size dependency.

[Table 1](#) presents the classification scheme developed for changeovers. Each changeover belongs to a certain class or combination of classes (B, T, and/or V), has some general characteristics, as well as characteristics related to the changeover matrix in terms of time and/or costs. The classification scheme characterizes changeovers on single machines. A scheduling problem will often include several machines that can each have multiple changeovers of different classes. This also means that even though a potential scarcity of machine capacities or of changeover resources (e.g., human operators) is an important consideration in scheduling problems (cf. [Tempelmeier & Buschkühl, 2008](#); [Tempelmeier & Copil, 2015](#)), it is not considered in the classification of individual changeovers.

2.1. Changeover classes

Three changeover classes are identified in [Table 1](#). A batch-dependent changeover (B) may be required when switching between batches. The changeovers can also depend on time or production volume. Time-dependent changeovers (T) are, for example, common in the food industry, in which machines are frequently cleaned at the end of the day to meet hygiene regulations. For volume-dependent changeovers (V), the changeover requirements depend on the produced volume or the frequency of use. These are, for example, often scheduled in the chemical industry after a specific number of batches to prevent buildup of impurities in processing items. Combined time and volume dependency of changeovers is also possible. For example, fouling of pipelines in the process industries could depend on both the time and the volume of product pumped.

2.2. General changeover characteristics

[Fig. 1](#) shows a graphical representation of general changeover characteristics.

Changeovers are classified as inseparable or separable based on their *separability*. Inseparable changeovers require the machine to be idle during the changeover, which is typical for the process industries. Separable changeovers, on the other hand, are prepared offline. These are common in discrete manufacturing. For instance, machine tools are often prepared offline in electronics assembly. However, they also arise in the process industries. An example is cheese production in which cheese forms are cleaned offline. For the just-in-time tool single-minute exchange of die (SMED), inseparable and separable (or internal and external) changeovers are distinguished and as much of the changeover operations as possible are treated as separable while running the machines ([Shingo, 1985](#)).

All changeovers are classified based on their *substitutability*. Changeovers may be insubstitutable, i.e., the changeover must take place and may not be substituted by any other changeover. A hierarchical structure may also be imposed on changeovers, leading to potential changeover substitutions. A matrix containing all of the changeovers can represent the changeover hierarchy. Changeovers may be substitutable within a class. For example, a time-dependent changeover that is scheduled every second hour may be substituted with a daily time-dependent changeover. Furthermore, changeovers may be substitutable across classes. For

Table 1
Classification scheme for changeovers.

Characteristic	Abbreviation	Potential value	Changeover class		
			Batch (B)	Time (T)	Volume (V)
General					
<u>S</u> eparability	{Sep}	• <u>i</u>nseparable (insep) • <u>s</u>eparable (sep)	X	X	X
<u>S</u> ubstitutability	{Sub}	• <u>i</u>nsubstitutable (insub) • <u>w</u>ithin class (wc) • <u>a</u>cross classes (ac) • <u>w</u>ithin and across classes (wac)	X	X	X
<u>R</u> eference point	{Ref}	• <u>s</u>tart/finish of batch (sfb) • <u>s</u>tart of production (sop) • <u>l</u>ast changeover (lc) • <u>f</u>ixed time point (ftp)	X	X	-
<u>F</u> lexibility	{Flex}	• <u>e</u>xact (ex) • <u>m</u>aximum (max) • <u>t</u>ime window (tw)	X	X	X
Changeover matrix: time/costs					
<u>P</u> roduct dependency	{Prod}	• <u>p</u>roduct(s) independent (pi) • <u>p</u>redecessor <u>p</u>roduct dependent (ppd) family (f)/ no family (-f) • <u>s</u>uccessor <u>p</u>roduct <u>d</u>ependent (spd) family (f)/ no family (-f) • <u>p</u>roduct <u>s</u>equences <u>d</u>ependent (seqd) family (f)/ no family (-f) natural seq. (ns)/ no natural seq. (-ns) triangular inequality kept (Δ)/ violated ($-\Delta$)	X	X	X
<u>B</u> atch-size dependency	{Size}	• <u>p</u>redecessor <u>p</u>roducts <u>h</u>istory (pph) • <u>b</u>atch-size(s) independent (bi) • <u>p</u>redecessor <u>b</u>atch-size <u>d</u>ependent (pbd) • <u>s</u>uccessor <u>b</u>atch-size <u>d</u>ependent (sbd) • <u>p</u>red. and <u>s</u>ucc. batch-size dependent (psbd) • <u>p</u>redecessor <u>b</u>atch-sizes <u>h</u>istory (pbh)	X	X	X

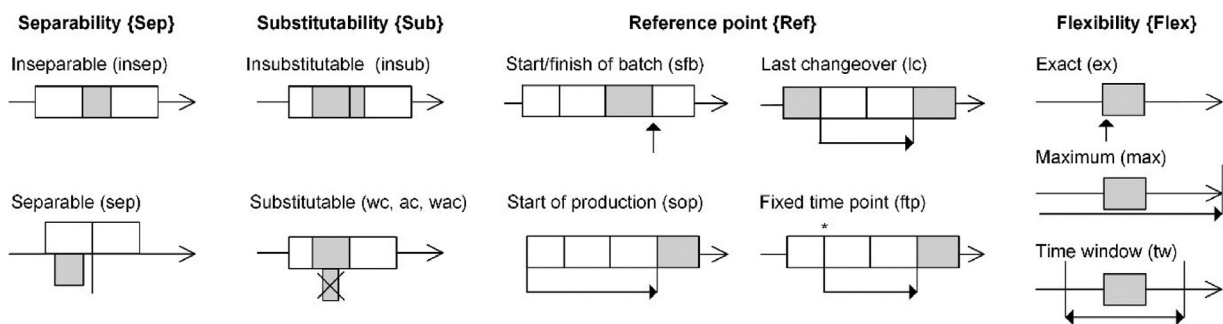


Fig. 1. Representation of general changeover characteristics.

example, in the food industry, a less intensive batch-dependent cleaning may be substituted with an intensive time-dependent cleaning. Finally, changeovers may be substitutable within and across classes, e.g., a time dependent changeover may be substituted with another time-dependent changeover and also with a volume-dependent changeover.

The starting time of a changeover is based on a *reference point*. The starting time for batch-dependent changeovers is always based on the start or finish of a batch. For time- and volume-dependent changeovers on the other hand, the starting time is based on start of production, start of last changeover, or on a fixed time point. The fixed time point is for example the start of planning horizon. The start of production and start of planning horizon are the same for problems with makespan minimization.

All changeovers are also classified based on the *flexibility* of their starting time. Changeovers may require an exact starting time with no flexibility. These are, for example, representative for food production, in which time-dependent cleanings take place at the end of each production day. Changeovers may also be more flexible, i.e., there may be an upper bound on the starting time (maximum) or both upper and lower bounds (time window).

2.3. Changeover-matrix characteristics

Fig. 2 shows a graphical representation of the characteristics related to the structure of the changeover matrix. The changeover-matrix characteristics can both be in terms of time and/or costs. The time and costs required to set up or clean a machine are known as the changeover time and costs, respectively. They do not necessarily have the same changeover-matrix characteristics.

Depending on their *product dependency*, changeovers may be independent of the produced product(s) as is frequently the case for time- and volume-dependent changeovers, but also for batch-dependent changeovers. Batch-dependent changeovers may also depend on the predecessor or successor batch. Family structure is a property of predecessor- and successor-dependent changeovers. Changeovers are minor when switching between products of the same family and major when switching between families. Furthermore, batch-dependent changeovers may be product sequence dependent in which case they depend on both the predecessor and successor batch. Sequence dependency is one of the basic characteristics of the process industries (Kallrath, 2002). Yoghurt production is a good example of a production system with

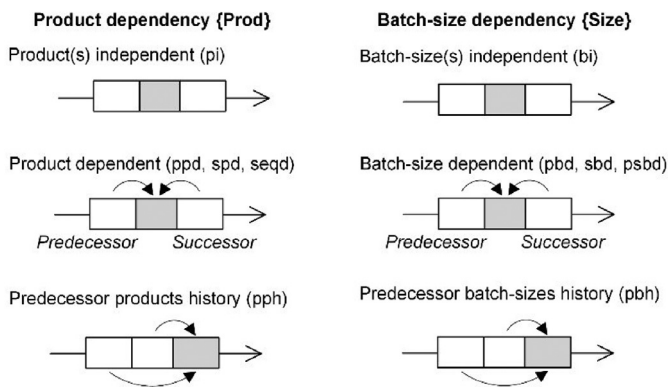


Fig. 2. Representation of changeover-matrix-related characteristics.

sequence dependency. When switching between products with different colors or flavors, cleaning techniques like cleaning, rinsing, or pushing depend on the preceding and succeeding products (Berlin, Sonesson, & Tillman, 2007). For example, an intensive cleaning technique is required when dark colored products precede light colored ones. Sequence-dependent changeovers have three properties. Family structure, the first property, has been described. Natural sequence between products, such as brighter to darker color, natural to stronger taste, or weaker to stronger microbiological contamination, is the second property of sequence dependency. Sorting products based on a natural sequence therefore reduces the required cleaning processes. Block planning, for example, exploits family structures and natural sequences of products for a simplified modeling approach (e.g., Günther, van Beek, Grunow, Lütke Entrup, & Zhang, 2006; Lütke Entrup, Günther, van Beek, Grunow, & Seiler, 2005). The third property of sequence dependency is triangular inequality. Triangular inequality is retained for most production environments. In this case, the changeover time between any two products is always less than that when a third product is produced between the two. For non-triangular changeovers, an intermediate product (often of lower grade) is produced between certain products to absorb contamination (Menezes, Clark, & Almada-Lobo, 2011). A minimum lot size is usually required for the intermediate product. Production of animal nutrition is a good example in which non-triangular cleanings apply (Clark, Morabito, & Toso, 2010). Claassen, Gerdessen, Hendrix, and van der Vorst (2016) also show that the optimal production schedule can vary significantly when including non-triangular changeovers. Finally, time- and volume-dependent changeovers may also depend on the history of the products being produced. For example, in meat processing, the knife wear may depend on the product types being produced.

Changeovers may be batch-size(s) independent or dependent depending on their *batch-size dependency*. For batch-size-independent changeovers, changeover duration does not depend on the batch size. For batch-size-dependent changeovers, changeover duration depends on the batch size of the predecessor, successor, predecessor and successor, or the predecessor batch-sizes history. For example for a large successor lot, more packaging material must be prepared leading to a longer setup time.

2.4. Classification

All changeovers can be classified according to the classification in Table 1, i.e., based on their class (B, T and/or V), their general characteristics (*Sep/Sub/Ref/Flex*), as well as changeover-matrix characteristics with regard to time $t(Prod/Size)$ and costs $c(Prod/Size)$. The classification of a changeover then takes the form

Class: (*Sep/Sub/Ref/Flex*); $t(Prod/Size)$; $c(Prod/Size)$.

Next, we illustrate the classification with two examples from the food industry. In this industry, changeovers often take the form of time-dependent cleanings done on the machines at the end of a production day. Assuming that this cleaning is substitutable across classes, with product- and batch-size-independent changeover matrix defined for times, its classification takes the form $T; (insep/ac/ftp/ex); t(pi/bi); c(-/-)$. The second example concerns a typical changeover from one product to another on a production line. This is often an inseparable and insubstitutable sequence-dependent batch changeover. If we assume a changeover matrix defined for costs, without a family structure or natural sequence, in which triangular inequality is maintained, and which is batch-size independent, then this changeover is classified as $B; (insep/insub/sfb/max); t(-/-); c(seqd(-/-ns;\Delta)/bi)$.

3. Related literature

The developed classification scheme can be used to facilitate the representation of changeovers when modeling lot sizing and scheduling problems. Our paper focuses on classifying and modeling changeovers with a particular focus on cleanings. Therefore, we have organized our literature review to first cover flowshops, the production environment, in which cleanings are most relevant. Due to their particular relevance for the case, we focus on no-wait flowshops. The second part reviews papers, which cover several changeover classes.

3.1. Simultaneous lot sizing and scheduling of flowshops

Lot sizing and scheduling problems have been extensively researched in the academic literature both independently and together. Lot sizing addresses timing and production amounts for satisfying demand over a time horizon (see the review by Buschkühl, Sahling, Helber, & Tempelmeier, 2010). Scheduling addresses sequencing of production activities (see the process industry reviews by Kallrath, 2002; Méndez, Cerdá, Grossmann, Harjunkski, & Fahl, 2006). As a result of sequence-dependent changeover times, different classes of changeovers, due dates, demand fluctuations, and tight capacities, lot sizing and scheduling problems are strongly interrelated and must be made simultaneously.

Two research communities in particular have developed mathematical models for simultaneous lot sizing and scheduling: the Operations Research (OR) community and the Process Systems Engineering (PSE) community (Amorim, Pinto-Varela, Almada-Lobo, & Barbósa-Póvoa, 2013). Approaches can be classified by their time representation into discrete- and continuous-time models. For discrete-time representation, all activities are scheduled on a discrete time scale in which the planning horizon is divided into a finite number of time intervals. The discrete-time formulations from the OR community are clustered into small and big bucket formulations, both aimed at minimizing inventory and setup costs (see the review by Copil, Wörbelauer, Meyr, & Tempelmeier, 2017). For continuous-time representation, all activities are scheduled accurately on a continuous time scale. There are also mixed models, which combine continuous- and discrete-time formulations. The block planning formulation is one of these. The choice of an appropriate time representation is a key issue, which must be addressed with the specific characteristics of the production system in mind.

Flowshop scheduling has received much attention in literature (see e.g., the reviews by Gupta & Stafford, 2006; Hejazi & Saghafian, 2005). However, to the best of our knowledge, lot sizing and scheduling of flowshops with a systematic analyses of changeovers has not been investigated in the literature. No-wait flowshops, which form an important cluster of flowshops, have

been studied extensively over the past few decades (see the reviews by Allahverdi, 2016; Hall & Sriskandarajah, 1996). However, only a few studies have addressed simultaneous lot sizing and scheduling of no-wait flowshops. The studies by Lin and Cheng (2001), Sriskandarajah and Wagneur (1999), and Wang and Cheng (2006) develop solution methods for the simultaneous lot sizing (batching) and scheduling of two-machine no-wait flowshops with sequence-independent setup times. Furthermore, Kumar, Bagchi, and Sriskandarajah (2000) and Kim and Jeong (2009) develop genetic algorithms for m -machine no-wait flowshops with sequence-independent setups. Also, Hall, Laporte, Selvarajah, and Sriskandarajah (2003) develop a dynamic programming algorithm for lot sizing and scheduling of m -machine no-wait flowshops with sequence-dependent setup times. All the analyzed studies minimize the makespan or maximum lateness, because time is commonly the principal criteria for no-wait flowshops. Furthermore, all of the studies only include one class of changeover and therefore none of the studies systematically addresses different changeovers.

3.2. Lot sizing and scheduling involving several changeover classes

In the food industry, scheduling of setups and especially cleanings plays a central role due to extensive hygienic requirements. In the following, we therefore focus on lot sizing and scheduling studies from the food industry in which two or more classes of changeovers are modeled. Table 2 presents an overview of the studies based on their application, method, time representation, objective, and changeover classification. Studies of integrated production scheduling and distribution are excluded.

Most of the reviewed papers are from the dairy industry, as there is an increasing interest in investigating changeovers in scheduling approaches from this sector. All papers apply MILP models, and Kilic (2011) also applies constraint programming. Furthermore, two approaches are based on block planning (see the papers by Günther et al., 2006; Kilic, 2011). All of the papers address lot sizing and scheduling; however in Kilic (2011) and Kopanos, Puigjaner, and Georgiadis (2012), it is not carried out simultaneously. Rather, lot sizing is carried out a priori. Either continuous- or mixed-time formulations are applied in the models. The objectives of the models include cost minimization, contribution margin maximization, and makespan minimization.

The changeover classification of the analyzed studies is described next. All of the papers study two changeover classes, i.e., batch- and time-dependent changeovers. Only our paper analyzes volume-dependent changeovers. All of the papers focus on inseparable cleanings, because inseparability is common for cleanings. The studies by Doganis and Sarimveis (2007), (2008) and Kopanos, Puigjaner, and Georgiadis, (2010), (2011) include substitution across classes. However, only time-dependent changeovers with exact starting times (like daily setup/shutdown) substitute batch-dependent changeovers, assuming that the equipment is clean when production is started on a new day. In our study, we incorporate substitution across classes that depends on the changeover duration. One other study on changeover substitution is found in the literature, see Gellert, Höhn, and Möhring (2011). They develop a genetic algorithm for scheduling yoghurt production and analyze changeover substitution across batch and time classes. However, this study does not address lot sizing and is therefore excluded from the literature overview.

The batch-dependent changeovers are almost all sequence dependent with diverse changeover-matrix structures. The models that minimize costs include changeover-matrix characteristics related to both time and costs. Many studies analyze cases in which there is a natural sequence, or both natural sequence and family structure, among the products. All studies, except for ours, address only batch-size-independent changeovers. Kopanos

et al. (2012) and Van Elzakker, Zondervan, Raikar, Grossmann, and Bongers (2012) each examine two separate batch-dependent changeovers: at the processing and packaging stages.

Time-dependent changeovers are analyzed in all of the papers. Most studies employ a reference point based on a fixed time point. In Van Elzakker et al. (2012), however, the reference point is based on start of production and in Kilic (2011) it is based on last changeover. Only our study addresses several reference points within the same changeover class, i.e., reference points based on a fixed time point and based on time since last changeover are modeled for time-dependent changeovers. With regard to flexibility, only our paper and the papers by Van Elzakker et al. (2012) and Kilic (2011) study time-dependent changeovers that are scheduled within a specific time window or a maximum upper bound. On the other hand, time-dependent changeovers with exact starting times are analyzed in many papers by reducing the daily production time by the changeovers time (for example, see the papers by Doganis & Sarimveis, 2007, 2008). Kopanos et al. (2010), (2011) study two separate time-dependent changeovers for a yoghurt packaging line including daily setup and daily shutdown.

There is a clear gap in the literature for a generic modeling approach addressing changeovers based on a systematic classification. As a consequence, none of the previous studies deals with (i) more than two changeover classes, (ii) volume-dependent changeovers, (iii) substitution of changeovers other than an exact time-dependent changeover substituting a batch-dependent changeover, or (iv) batch-size dependency of changeovers. Our study, in contrast, represents all changeover classes and different combinations of their characteristics in a simultaneous lot sizing and scheduling model.

4. Lot sizing and scheduling of flowshops

4.1. Model overview

We develop a generic MILP model for lot sizing and scheduling of two-machine flowshops. The model provides the core structure for modeling changeovers. When the model is applied in a specific context, it will often be necessary to extend the model such that other aspects than changeovers are included. This can easily be done because of the adaptable and declarative nature of the approach.

Lot sizing and scheduling is conventionally studied using models that trade-off between inventory and setup costs. However, minimizing inventory and setup costs is only suitable if the costs can be identified as out-of-pocket costs for the individual products (Günther, 2014). Furthermore, inventory costs are often low for scheduling problems that typically have short planning horizons, and personnel costs often dominate setup costs. Hence, our objective is to minimize the makespan over a short horizon. Here, we also avoid idle times on machines, which occasion additional cleanings. A sequence-based, continuous-time representation is selected to allow precise scheduling of all activities and due to its flexibility in incorporating different changeovers. Discrete-time formulations are often created in a relatively straightforward manner; however, they are difficult to use in many industrial projects as they often result in large combinatorial problems when selecting sufficiently small intervals.

Deterministic demand volumes of products must be fulfilled. The model determines the number, sequence, size, and starting times of lots as well as the changeovers. The number of potential lots per product is predetermined and may vary for different products. In practice, the number of potential lots per product can, for example, be determined as the maximum number of lots, over which the demand volume will be divided considering the changeover time and other influencing factors. The lots, which

Table 2
Overview of changeovers in the literature on lot sizing and scheduling in the food industry involving several changeover classes.

Publication	Application	Method	Time repr.	Objective	Class	Changeover classification		
						General characteristics (Sep/Sub/Ref/Flex)	Changeover-matrix characteristics Time(Prod/Size)	Changeover-matrix characteristics Costs(Prod/Size)
Kopanos et al. (2012)	Ice-cream	MILP	Cont.	Min make span	B; B; T; T;	(insep/insub/sfb/max); (insep/insub/sfb/max); (insep/insub/ftp/ex); (insep/insub/ftp/ex);	t(seqd(-f;ns; Δ)/bi); t(seqd(-f;ns; Δ)/bi); t(pi/bi); t(pi/bi);	c(-/-) c(-/-) c(-/-) c(-/-)
Van Elzakker et al. (2012)	Ice-cream	MILP	Cont.	Min make span	B; B; T; T;	(insep/insub/sfb/max); (insep/insub/sfb/max); (insep/insub/sop/tw); (insep/insub/sop/tw);	t(seqd(-f;-ns; Δ)/bi); t(seqd(-f;-ns; Δ)/bi); t(pi/bi); t(pi/bi);	c(-/-) c(-/-) c(-/-) c(-/-)
Doganis and Sarimveis (2007)	Yoghurt	MILP	Mix	Min costs	B; T;	(insep/ac/sfb/max); (insep/ac/ftp/ex);	t(seqd(-f;ns; Δ)/bi); t(pi/bi);	c(seqd(-f;ns; Δ)/bi) c(-/-)
Doganis and Sarimveis (2008)	Yoghurt	MILP	Mix	Min costs	B; T;	(insep/ac/sfb/max); (insep/ac/ftp/ex);	t(seqd(-f;ns; Δ)/bi); t(pi/bi);	c(seqd(-f;ns; Δ)/bi) c(-/-)
Kopanos et al. (2010)	Yoghurt	MILP	Mix	Min costs	B; T;	(insep/ac/sfb/max); (insep/ac/ftp/ex);	t(seqd(f;ns; Δ)/bi); t(pi/bi);	c(seqd(f;ns; Δ)/bi) c(-/-)
Kopanos et al. (2011)	Yoghurt	MILP	Mix	Min costs	B; B; T; T;	(insep/insub/sfb/max); (insep/insub/sfb/max); (insep/insub/ftp/ex); (insep/insub/ftp/ex);	t(seqd(f;ns; Δ)/bi); t(spd(-f)/bi); t(pi/bi); t(pi/bi);	c(seqd(f;ns; Δ)/bi) c(-/-) c(-/-) c(-/-)
Kilic (2011)	Milk	MILP, constr. progr.	Cont.	Min make span	B; T; T;	(insep/insub/sfb/max); (insep/insub/lc/max); (insep/insub/lc/max);	t(seqd(f;ns; Δ)/bi); t(pi/bi); t(pi/bi);	c(-/-) c(-/-) c(-/-)
Günther et al. (2006)	Scalded sausage	MILP	Mix	Max margin	B; T;	(insep/insub/sfb/max); (insep/insub/ftp/ex);	t(seqd(f;ns; Δ)/bi); t(pi/bi);	c(-/-) c(-/-)
This paper (2016)	Cheese	MILP	Cont.	Min make-span	B; T; V;	(insep/insub/sfb/max); (insep/*/*/*); (insep/*/*/*);	t(*/*/*); t(pi/bi); t(pi/bi);	c(-/-) c(-/-) c(-/-)

*) Different combinations of characteristics are analyzed.

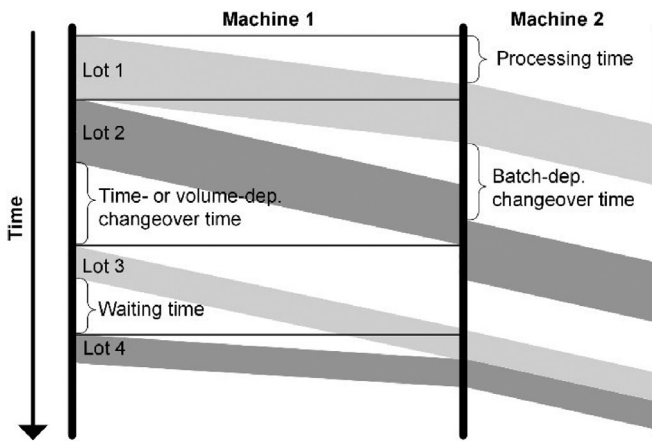


Fig. 3. Representation of a two-machine flowshop.

are moved through the system without overtaking each other, have heterogeneous processing times, precedence relationships, and due dates. Products are first loaded onto the first machine in lots and then processed there. Afterwards they are loaded onto the next machine.

The model addresses all of the changeover classes in Table 1, i.e., those containing batch-, time-, and volume-dependent changeovers. The focus is on the time dimension of the changeover-matrix characteristics, because of the makespan objective. Changeover resources are assumed to be uncapacitated, because the focus is on primary operations. All changeover classes are inseparable, which is common for this type of production environment. Substitution of changeovers on the same machine is incorporated across classes, depending on the changeover duration, meaning that shorter changeovers are substituted with longer ones.

Batch-dependent changeovers, $B; (insep/insub/sfb/max); t(seqd(-f;-ns;\Delta)/bi); c(-/-)$, are required on the second machine. This is typical for many process industry problems such as those in which continuous volumes are transferred to discrete and format change is required in the second stage. The reference point for the batch-dependent changeovers is based on the start/finish of last batch and the starting time is within an upper bound (maximum). The model is developed for sequence-dependent changeovers, which is the most complicated case, so other product dependencies can be incorporated. A general changeover-matrix structure in terms of family, natural sequence, and triangular inequality is modeled. The model is developed for batch-size-independent changeovers. Model adjustments for batch-size-dependent changeovers, $B; (insep/insub/sfb/max); t(spd(-f)/sbd); c(-/-)$, are also shown.

We assume that there are time-, $T; (insep/ac/ftp/tw); t(pi/bi); c(-/-)$, and volume-dependent changeovers, $V; (insep/ac/ftp/tw); t(pi/bi); c(-/-)$, on the first machine. The changeovers occur within a specific time window, thereby exact and maximum criteria (upper bound) are also incorporated. An alternative formulation for exact time-dependent changeovers is also shown, $T; (insep/ac/ftp/ex); t(pi/bi); c(-/-)$. The reference point for the time- and volume-dependent changeovers is based on the start of planning horizon (fixed time point). For makespan minimization the start of planning horizon is the same as the start of production. How time-dependent changeovers can be modeled based on the time since last changeover, $T; (insep/ac/lc/max); t(pi/bi); c(-/-)$, is also shown. Fig. 3 shows the connection between the product speed and changeovers for the two-machine configuration.

4.2. Model formulation

The following notation is introduced for the problem:

Indices and index sets

$p \in P$	product
$l \in L$	lot
$l \in L(p)$	lot belonging to product p
$k \in K$	time-dependent changeover
$j \in J$	volume-dependent changeover

Parameters

α_p	demand for product p [units]
β_l	minimum lot size of lot l [units]
γ_l	due date of lot l [time units]
$\delta_{ll'}$	= 1 if lot l must be produced after lot l' (0, otherwise)
ε	loading duration of one unit [time units]
ζ_l	processing time of lot l on machine 1 [time units]
η	lowest processing time of all lots on machine 1 [time units]
$\theta_{ll'}$	duration of sequence-dependent batch changeover between lot l and lot l' (l precedes l') [time units]
κ	duration of time-dependent changeover [time units]
$\lambda_k^{\min}$	earliest starting time of time-dependent changeover k —lower bound [time units]
$\lambda_k^{\max}$	latest starting time of time-dependent changeover k —upper bound [time units]
μ	minimum number of time-dependent changeovers [changeovers]
ν	changeover interval for time-dependent changeovers [time units]
ξ	duration of volume-dependent changeover [time units]
$\sigma_j^{\min}$	earliest starting of volume-dependent changeover j —lower bound [units]
$\sigma_j^{\max}$	latest starting of volume-dependent changeover j —upper bound [units]
π	minimum number of volume-dependent changeovers [changeovers]
ρ	changeover interval for volume-dependent changeovers [units]
σ_l	batch-size coefficient of lot l
M_1, M_2	sufficiently large positive values

Decision variables

$C_{\max}$	makespan [time units]
X_l	amount produced in lot l [units]
$S1_l$	starting time of loading lot l on machine 1 [time units]
$C1_l$	completion time of loading lot l on machine 1 [time units]
$S2_l$	starting time of loading lot l on machine 2 [time units]
$C2_l$	completion time of loading lot l on machine 2 [time units]
E_l	duration of batch-dependent changeover before lot l [time units]
N_k	starting time of time-dependent changeover k [time units]
Q_l	cumulated production amount after finishing lot l [units]
$Y_{ll'}$	= 1 if lot l processed before lot l' (0, otherwise)
Z_{lk}	= 1 if lot l scheduled immediately before time-dependent changeover k (0, otherwise)
H_{lk}	= 1 if lot l scheduled before exact time-dependent changeover k (0, otherwise)
A_k	= 1 if time-dependent changeover k is scheduled last (0, otherwise)
V_{ij}	= 1 if lot l scheduled immediately before volume-dependent changeover j (0, otherwise)

The lot sizing and scheduling model is formulated as follows:

Objective function

The objective is to minimize the makespan (1).

$$\text{Min } C_{\max} \quad (1)$$

subject to

Demand, lot size, makespan, due dates, and precedence relationships

Demand for a product must be fulfilled (2). A minimum lot size is introduced with constraints (3). If the triangular inequality is violated, then these constraints impose sufficient intermediate product lot sizes to absorb all contamination. This is generally modeled so that a minimum lot size is imposed on all lots and not just on the intermediate lot. Completion time of the last lot on the second machine determines the makespan (4). All lots belonging to the same product must be finished before the respective due date of the product (5), assuming this is feasible for all problem instances. Due dates are based on customer or downstream processing requirements, reflected for example in instructions from higher planning levels. If due dates are not available for some products, they are set to the end of the planning horizon. Constraints (6) model the precedence relationships among products. Alternatively, a large sequence-dependent batch changeover time can be set for these products.

$$\sum_{l \in L(p)} X_l \geq \alpha_p \quad \forall p \in P \quad (2)$$

$$X_l \geq \beta_l \quad \forall l \in L \quad (3)$$

$$C_{\max} \geq C2_l \quad \forall l \in L \quad (4)$$

$$C2_l \leq \gamma_p \quad \forall p \in P, l \in L(p) \quad (5)$$

$$S1_{l'} \cdot \delta_{ll'} \leq S1_l \quad \forall l, l' \in L \quad (6)$$

Timing on machines

Constraints (7) define the completion times for loading on machine 1. Note that products are subsequently processed on machine 1 before arriving at machine 2. It is assumed that no changeovers occur before loading the first lot. Constraints (8) set the starting times for loading on machine 2. Constraints (9) define the completion times on machine 2.

$$C1_l = S1_l + \varepsilon \cdot X_l \quad \forall l \in L \quad (7)$$

$$S2_l \geq S1_l + \varepsilon + \zeta_l \quad \forall l \in L \quad (8)$$

$$C2_l = S2_l + \varepsilon \cdot X_l \quad \forall l \in L \quad (9)$$

Constraints (10, 11) define the starting times on machine 1, accounting for the waiting times resulting from differences in processing times. If negative values result, constraints (15, 16) prevent lots from starting earlier than the completion time of the previous lot. Constraints (10) apply if lot l' is produced after lot l , and constraints (11) if lot l is produced after lot l' . The constant M_1 represents the length of the planning horizon in time units. The variable $Y_{ll'}$ is always zero when $l = l'$ (12).

$$S1_{l'} \geq C1_l - (\zeta_{l'} - \zeta_l) - M_1 \cdot (1 - Y_{ll'}) \quad \forall l, l' \in L : (l \neq l') \quad (10)$$

$$S1_l \geq C1_{l'} - (\zeta_l - \zeta_{l'}) - M_1 \cdot Y_{ll'} \quad \forall l, l' \in L : (l \neq l') \quad (11)$$

$$Y_{ll'} = 0 \quad \forall l, l' \in L : (l = l') \quad (12)$$

Constraints for all changeover classes are formulated next.

Batch-dependent changeovers

Constraints (13, 14) define the starting times on machine 2, accounting for the duration of sequence-dependent batch changeovers. The batch-dependent changeovers can be changed to predecessor- or successor-dependent changeovers by changing the parameter $\theta_{ll'}$.

$$S2_{l'} \geq C2_l + \theta_{ll'} - M_1 \cdot (1 - Y_{ll'}) \quad \forall l, l' \in L : (l \neq l') \quad (13)$$

$$S2_l \geq C2_{l'} + \theta_{l'l} - M_1 \cdot Y_{ll'} \quad \forall l, l' \in L : (l \neq l') \quad (14)$$

The modeling of the batch-dependent changeovers can also be modified to include batch-size dependency. We show this for changeovers dependent on successor product and successor batch size. Predecessor batch-size-dependent changeovers can be modeled analogously. A continuous variable E_l , which defines the duration of changeover before lot l , is utilized. In constraints (13), the parameter $\theta_{ll'}$ is replaced by $E_{l'}$, and in constraints (14) the parameter $\theta_{l'l}$ is replaced by E_l . Furthermore, constraints (A1) are added to set value on the new variable. The duration of the changeover depends on the batch size of the lot and a batch-size coefficient (σ_l).

$$X_l \cdot \sigma_l \leq E_l \quad \forall l \in L \quad (A1)$$

Time-dependent changeovers

If a time-dependent changeover is scheduled on machine 1 (15, 16) and it is larger than the processing-time differences in constraints (10, 11), then the downtime on machine 1 must be larger or equal to the duration of the changeover.

$$S1_{l'} \geq C1_l + \kappa \cdot Z_{lk} - M_1 \cdot (1 - Y_{ll'}) \quad \forall l, l' \in L : (l \neq l'), k \in K \quad (15)$$

$$S1_l \geq C1_{l'} + \kappa \cdot Z_{l'k} - M_1 \cdot Y_{ll'} \quad \forall l, l' \in L : (l \neq l'), k \in K \quad (16)$$

Constraints (17) determine the number of time-dependent changeovers, based on a fixed reference point (since start of planning horizon). Constraints (18) set the order of time-dependent changeovers (e.g., if second changeover is scheduled, then first changeover must be scheduled).

$$\sum_{l \in L} \sum_{k \in K} Z_{lk} \geq \frac{C_{\max} - \lambda_1^{\max}}{\nu} \quad (17)$$

$$\sum_{l \in L} Z_{lk} \geq \sum_{l \in L} Z_{lk'} \quad \forall k, k' \in K : (k < k') \quad (18)$$

The time-dependent changeovers are flexible, starting within their respective time windows (19, 20). Also, a value is set on the binary variable that indicates if changeover precedes a specific lot.

$$C1_l \leq \lambda_k^{\max} + M_1 \cdot (1 - Z_{lk}) \quad \forall l \in L, k \in K \quad (19)$$

$$C1_l \geq \lambda_k^{\min} - M_1 \cdot (1 - Z_{lk}) \quad \forall l \in L, k \in K \quad (20)$$

For time-dependent changeovers with only an upper bound (maximum), only constraints (19) are imposed. The upper and lower bounds are equal for exact time-dependent changeovers. Alternatively, exact time-dependent changeovers can be modeled by substituting constraints (15–20) with constraints (B1, B2).

$$C1_l \leq \lambda_k^{\min} + M_1 \cdot (1 - H_{lk}) \quad \forall l \in L, k \in K \quad (B1)$$

$$S1_l \geq \lambda_k^{\min} + \kappa - M_1 \cdot H_{lk} \quad \forall l \in L, k \in K \quad (B2)$$

In the following, we show an alternative formulation for time-dependent changeovers that are not based on time since start of planning horizon as a reference point, but rather time since last changeover with a maximum upper bound on the starting time

(flexibility). Other flexibility characteristics can also be incorporated. Constraints (17, 19, 20) are substituted with constraints (C1–C7). The changeover frequency is set with constraints (C1–C3). Only one changeover is the last changeover (C4). Constraints (C5–C7) set the starting times of the changeovers and a value on the binary variable that indicates if changeover precedes a specific lot, similarly as in constraints (19, 20).

$$N_1 \leq \lambda_1^{\max} \quad (C1)$$

$$N_{k+1} \leq N_k + v \quad \forall k \in K : (k < |K|) \quad (C2)$$

$$C_{\max} \leq N_k + v + M_1 \cdot (1 - A_k) \quad \forall k \in K \quad (C3)$$

$$\sum_{k \in K} A_k = 1 \quad (C4)$$

$$N_k \leq M_1 \cdot \sum_{l \in L} Z_{lk} \quad \forall k \in K \quad (C5)$$

$$C1_l \leq N_k + M_1 \cdot (1 - Z_{lk}) \quad \forall l \in L, k \in K \quad (C6)$$

$$C1_l \geq N_k - M_1 \cdot (1 - Z_{lk}) \quad \forall l \in L, k \in K \quad (C7)$$

Volume-dependent changeovers

Volume-dependent changeovers are scheduled with constraints (21, 22), similar to the way time-dependent changeovers are scheduled. Substitution can take place between time and volume-dependent changeovers.

$$S1_{l'} \geq C1_l + \xi \cdot V_{lj} - M_1 \cdot (1 - Y_{ll'}) \quad \forall l, l' \in L : (l \neq l'), j \in J \quad (21)$$

$$S1_l \geq C1_{l'} + \xi \cdot V_{l'j} - M_1 \cdot Y_{ll'} \quad \forall l, l' \in L : (l \neq l'), j \in J \quad (22)$$

Constraints (23) set the number of volume-dependent changeovers, based on a fixed reference point. Formulation similar to that described for the time-dependent changeovers (C1–C7) can be developed for a reference point depending on volume since last changeover, but adjusted for volumes. Constraints (24) set the order of volume-dependent changeovers.

$$\sum_{l \in L} \sum_{j \in J} V_{lj} \geq \sum_{p \in P} \frac{\alpha_p}{\rho} \quad (23)$$

$$\sum_{l \in L} V_{lj} \geq \sum_{l' \in L} V_{l'j'} \quad \forall j, j' \in J : (j < j') \quad (24)$$

The volume-dependent changeovers are flexible, i.e., scheduled within the upper and lower bounds (25, 26). For volume-dependent changeovers with only an upper bound (maximum), only constraints (25) are imposed. Upper and lower bounds are equal for exact volume-dependent changeovers.

$$Q_l \leq \sigma_j^{\max} + M_2 \cdot (1 - V_{lj}) \quad \forall l \in L, j \in J \quad (25)$$

$$Q_l \geq \sigma_j^{\min} - M_2 \cdot (1 - V_{lj}) \quad \forall l \in L, j \in J \quad (26)$$

Constraints (27, 28) set the cumulated production amounts. Constraints (29) enforce the cumulated amount of the first lot being greater than or equal to the production amount. Constraints (30) restrict the cumulated amount to be less than the total production amount.

$$Q_{l'} \geq Q_l + X_{l'} - M_2 \cdot (1 - Y_{ll'}) \quad \forall l, l' \in L : (l \neq l') \quad (27)$$

$$Q_l \geq Q_{l'} + X_l - M_2 \cdot Y_{ll'} \quad \forall l, l' \in L : (l \neq l') \quad (28)$$

$$X_l \leq Q_l \quad \forall l \in L \quad (29)$$

$$\sum_{l' \in L} X_{l'} \geq Q_l \quad \forall l \in L \quad (30)$$

Tightening constraints

Tightening constraints are introduced to reduce computational effort. A lower bound on the makespan similar to that in Kopanos et al. (2012) is introduced. The lower bound is calculated as the minimum processing time plus the net loading times plus a minimum duration of time- and volume-dependent changeovers (31). Waiting and sequence-dependent batch changeovers are not included, as they depend on the resulting schedule. Constraints (32) break the symmetries of lots of same product assuming that a lower indexed lot is produced first. Constraints (33) set the number of lots that immediately precede a changeover to at most one. Likewise, constraints (34) set the number of changeovers scheduled immediately after a lot to at most one.

$$C_{\max} \geq \eta + \sum_{p \in P} \varepsilon \cdot \alpha_p + \mu \cdot \kappa + \pi \cdot \xi \quad (31)$$

$$Y_{ll'} = 1 \quad \forall l, l' \in L(p) : (l < l') \quad (32)$$

$$\sum_{l \in L} Z_{lk} \leq 1 \quad \forall k \in K, \quad \sum_{l \in L} V_{lj} \leq 1 \quad \forall j \in J \quad (33)$$

$$\sum_{k \in K} Z_{lk} \leq 1 \quad \forall l \in L, \quad \sum_{j \in J} V_{lj} \leq 1 \quad \forall l \in L \quad (34)$$

Variable domains

Constraints (35, 36) define the variable domains of decision variables.

$$C_{\max} \geq 0, X_l \geq 0, S1_l \geq 0, C1_l \geq 0, S2_l \geq 0, C2_l \geq 0, E_l \geq 0, N_k \geq 0, Q_l \geq 0 \quad \forall l \in L, k \in K \quad (35)$$

$$Y_{ll'} \in \{0, 1\}, Z_{lk} \in \{0, 1\}, H_{lk} \in \{0, 1\}, A_k \in \{0, 1\}, V_{lj} \in \{0, 1\} \quad \forall l, l' \in L, k \in K, j \in J \quad (36)$$

The developed model comprehensively covers all changeover classes and their various characteristics. On machine 2, batch-dependent changeovers (13, 14) are scheduled between successive lots. The changeover interactions of time- and volume-dependent changeovers on machine 1 are modeled through substitution across classes. Different changeover classes can therefore be scheduled between successive lots on machine 1, i.e., time- (15, 16), or volume-dependent changeovers (21, 22). For insubstitutable time- and volume-dependent changeovers, separate constraints would not be required for each class because the changeover times are added up. Furthermore, for additional changeovers of a certain class, the corresponding sets of constraints, variables and parameters must be duplicated. In the case study below, two time-dependent changeovers are implemented.

5. Application to cheese production

We apply the developed model to cheese production. This application is selected, because cleanings play a major role in cheese production for health reasons and due to regulatory standards.

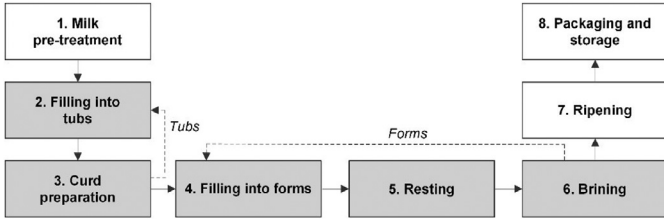


Fig. 4. Processing steps in soft and blue cheese production (steps 2 through 6 are analyzed in this study).

5.1. Production process

All cheese types can be roughly categorized into hard, semi-hard, soft, semi-soft, and cream cheeses, depending on moisture content. Each cheese type has its own special production process, ripening process, taste, and texture. The paper focuses on the production of soft and blue cheese (categorized as semi-soft cheese). Fig. 4 shows the processing steps followed by a short description of each step based on Bylund (2003), Walstra, Wouters, and Geurts (2010), and information from practice. Note that the term "filling" is used for cheese production as opposed to the term "loading" in the model description.

The raw milk is first standardized to around 3% fat content, pasteurized at approximately 72 °C, and cooled to 32 °C (1). Next, the milk is filled into tubs of uniform size (2). Rennet is then added. The rennet causes the casein in the milk to coagulate, which is a fundamental process in cheese production. The amount depends on the specific cheese type and recipe. Starter culture, or other additives, can also be added. The milk clots after mixing and the clotted milk is then cut to expel the whey from the curd (3). After curd preparation, the curd is filled into forms without waiting (4). The forms give the cheese its size and shape. Next, the cheese rests under special conditions for several hours (5). While resting, the remaining whey is dripped or pressed out to obtain the correct dry matter content. The cheese is also turned several times. After resting, the cheese is typically salted in a salt bath to increase its shelf life (6). The cheese is subsequently removed from the forms after which it ripens for several days under special conditions (7). Water losses and deterioration of texture should be avoided. After a few days, some cheeses are pierced to enable oxygen to enter so that blue mold can grow. The cheese is ultimately packaged and stored (8).

The main capacity-constrained processes constituting the flowshop are analyzed. Enough tubs are assumed to be available. However the use of different form types, steps 4 to 6, must be tracked as forms are blocked for several hours in a capacitated resting area after being filled and represent a key bottleneck. Addressing processing steps 2 and 3 is also necessary as the product sequence and timing of these steps influences scheduling of the following steps due to a no-wait condition between steps 3 and 4. No-wait conditions often apply in the food industry due to the characteristics of production technologies or of intermediate products. Fig. 5 shows a representation of machine 1 (coagulator) and machine 2 (filler), which constitute a two-machine no-wait flowshop. A no-wait condition is employed at the interfaces between the machines, whereby lots are immediately loaded onto the next machine. Sequence-dependent batch changeovers result on machine 2 due to exchanging of form types and cleaning of machines. Furthermore, time- and/or volume-dependent cleanings must be addressed as cleaning operations are very important in the production of cheeses. Tight coordination of the changeovers across the production stages is required due to the no-wait condition.

Despite increasing interest in lot sizing and scheduling in the dairy industry, the literature focuses mainly on yoghurt produc-

tion. Cheese production is distinct from yoghurt production, e.g., because products are filled in forms, and heterogeneous clotting and resting times must be accounted for. To the best of our knowledge, only the study by Claassen and van Beek (1993) examines cheese scheduling. However, this study focuses on packaging without addressing specific cheese production challenges such as cleanings, sequence dependency, and the capacity restrictions of forms.

5.2. Modeling no-wait condition and form usage

This section describes the adaptations of the developed approach for modeling the no-wait condition between the machines and for tracking form usage. Discrete time periods are used to track form usage within each time period. That the required number of forms for the whole lot is available when filling starts is assumed, because receiving additional forms after a lot is started is unrealistic from a practical perspective. The notation below is added to the model presented in Section 4.

Indices

$f \in F$	form type
$t \in T$	period

Parameters

τ	length of discrete time period [time units]
ν_l	number of forms filled from one tub for lot l [number of forms]
φ_l	resting and brining time of lot l on machine 2, and separable form cleaning time [periods]
χ_{lf}	= 1 if lot l requires form type f (0, otherwise)
ψ_f	number of available forms of form type f –capacity [number of forms]
M_3	sufficiently large positive value

Decision variables

B_{lt}	number of filled forms for lot l in period t [number of forms]
R_{lt}	number of released forms for lot l in period t [number of forms]
I_{ft}	inventory of form type f in period t [number of forms]
W_{lt}	= 1 if filling of forms for lot l carried out in period t (0, otherwise)
U_l	consumption period of forms for lot l [period]

No-wait condition

Constraints (8) are replaced with constraints (37), whereby lots are immediately loaded onto machine 2 after being processed on machine 1. Also, constraints (10, 11, 13, 14) are replaced with constraints (38, 39). The starting times on machine 1 must account for the duration of sequence-dependent batch changeovers and the waiting times resulting from processing-times differences. For example, lot 2 in Fig. 3 would have to start later on machine 1 due to the no-wait condition.

$$S2_l = S1_l + \varepsilon + \zeta_l \quad \forall l \in L \quad (37)$$

$$S1_{l'} \geq C1_l + (\theta_{ll'} - (\zeta_{l'} - \zeta_l)) - M_1 \cdot (1 - Y_{ll'}) \quad \forall l, l' \in L : (l \neq l') \quad (38)$$

$$S1_l \geq C1_{l'} + (\theta_{ll'} - (\zeta_l - \zeta_{l'})) - M_1 \cdot Y_{ll'} \quad \forall l, l' \in L : (l \neq l') \quad (39)$$

Form usage

In the sequel, continuous starting times for filling into forms are transformed into discrete time periods to track form usage. Constraints (40) set values on the discrete time periods when forms

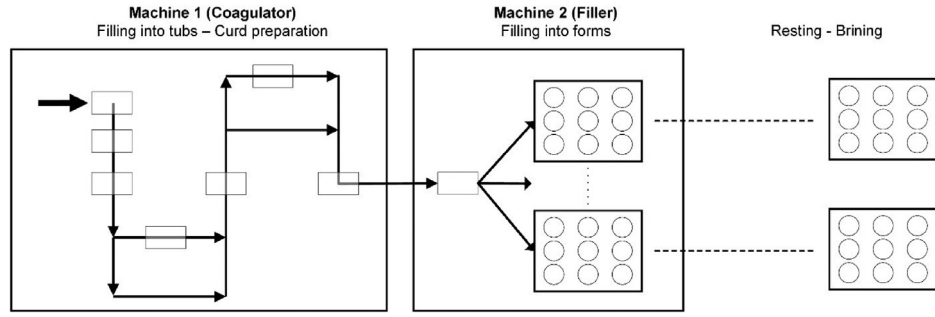


Fig. 5. Detailed view of the analyzed two-machine no-wait flowshop for cheese production.

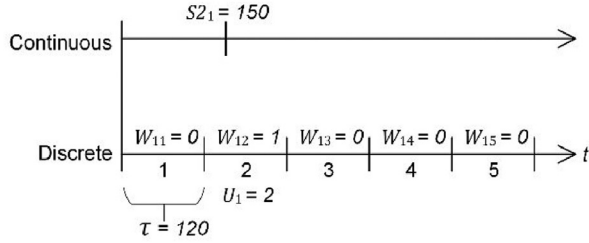


Fig. 6. Relationship between continuous- and discrete-time axes.

are consumed. Constraints (41) and (42) subsequently set the corresponding values for the binary variables that specify if filling of lot l is carried out in discrete time period t . Combined, constraints (40–42) thereby link the continuous time axis and the discrete time axis through decision variables $S2_l$, U_l , and W_{lt} . Fig. 6 illustrates the relationship between the time axes with an example. If the starting time for filling forms for lot 1 ($S2_1$) is at 150 time units and the length of the discrete time period is 120 time units, then the discrete time period in which forms are consumed (U_1) is set to 2. Furthermore, the binary variable W_{12} is assigned a value of 1.

$$\frac{S2_l}{\tau} \leq U_l < \frac{S2_l + \tau}{\tau} \quad \forall l \in L \quad (40)$$

$$U_l = t \cdot W_{lt} \quad \forall l \in L, t \in T \quad (41)$$

$$\sum_{t \in T} W_{lt} = 1 \quad \forall l \in L \quad (42)$$

At least the whole production lot is filled (43). Constraints (44) force the value of B_{lt} to zero when W_{lt} is zero. Constraints (45) determine the release time of forms by adding the resting time. Forms are not released unless they are first filled (46).

$$X_l \cdot v_l = \sum_{t \in T} B_{lt} \quad \forall l \in L \quad (43)$$

$$B_{lt} \leq M_3 \cdot W_{lt} \quad \forall l \in L, t \in T \quad (44)$$

$$B_{lt} = R_{l,t+\varphi_l} \quad \forall l \in L, t \in T : t \leq T - \varphi_l \quad (45)$$

$$R_{lt} = 0 \quad \forall l \in L, t \in T : t \leq \varphi_l \quad (46)$$

It is assumed that all forms are available at the beginning of the planning horizon, e.g., due to extensive cleaning at the end of the previous planning horizon. The inventory (number of blocked forms) in the first period equals the number of forms filled in this period (47). Constraints (48) model the inventory balances of the

other periods. The inventory of forms in each period must be less than the number of available forms (49).

$$I_{ft} = \sum_{l \in L} B_{lt} \cdot \chi_{lft} \quad \forall f \in F, t \in T : t = 1 \quad (47)$$

$$I_{ft} = I_{f,t-1} + \sum_{l \in L} B_{lt} \cdot \chi_{lft} - \sum_{l \in L} R_{lt} \cdot \chi_{lft} \quad \forall f \in F, t \in T : t \geq 2 \quad (48)$$

$$I_{ft} \leq \psi_f \quad \forall f \in F, t \in T \quad (49)$$

Variable domains

Constraints (50, 51) define the variable domains of decision variables.

$$B_{lt} \geq 0, R_{lt} \geq 0, I_{ft} \geq 0 \quad \forall l \in L, f \in F, t \in T \quad (50)$$

$$W_{lt} \in \{0, 1\}, U_l \geq 0 \text{ and int } \forall l \in L, t \in T \quad (51)$$

5.3. Case study data

The case study includes the production of 30 different products, in which the weekly demand volumes (α_p) are given. The number of possible lots per product is determined together with the cheese dairy based on the demand volumes. A minimum lot size (β_l) is not required. The objective is to minimize the makespan over a one week planning horizon such that production on weekends is minimized. Total clotting, cutting, and stirring times of products are 80–144 minutes (ζ_l). The tubs are all of the same size with filling time (ε) of 104 seconds. There are three different form types and the number of available forms is limited to 1000 to 4000 (ψ_f), depending on the type. There are between 5 and 13 forms per tub (v_l). Resting and brining times of products are 13–19 hours (φ_l) including the separable cleaning time of forms. The period length for tracking form usage is set to 2 hours (τ).

The required setups and cleanings are described next. First, there are batch-dependent changeovers on the second machine, $B; (insep/insub/sfb/max); t(seqd(-f;-ns;\Delta)/bi); c(-/-)$, in which the sequence-dependent duration ($\theta_{ll'}$) varies depending on whether products require the same form type (either 0 or 15 minutes cleaning) or different form types (15 minutes cleaning plus 15 minutes setup). Also, two time-dependent cleanings classified as $T; (insep/insub/ftp/tw); t(pi/bi); c(-/-)$ and $T; (insep/insub/ftp/ex); t(pi/bi); c(-/-)$ are considered on the first machine. The first one, denoted with superscript (1), is a time-window-based cleaning. It takes 42 minutes ($\kappa^{(1)}$) and is scheduled in the middle of each production day, i.e., between 8 ($\lambda_k^{min(1)}$) and 12 hours ($\lambda_k^{max(1)}$) of production. The cleaning interval is 24 hours ($\nu^{(1)}$). The second time-dependent cleaning, denoted with superscript (2), is an exact cleaning. It takes 4 hours ($\kappa^{(2)}$) and is scheduled at the end of each production day, i.e., after 20 hours ($\lambda_k^{min(2)}$) of production. Substitution within the time-dependent

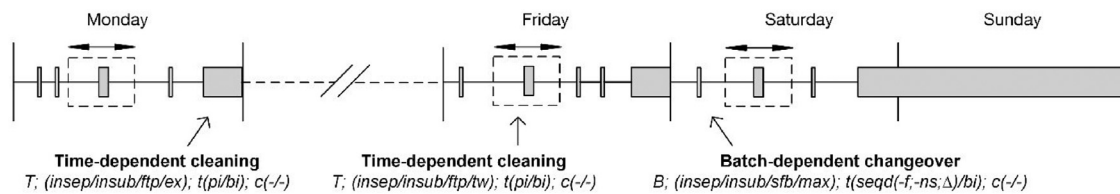


Fig. 7. Analyzed setups and cleanings for a one week planning horizon.

cleaning class will not occur, because cleanings are scheduled at different times during the day. Also, substitution across classes will not occur, because the classes are scheduled on different machines. The minimum number of cleanings ($\mu^{(1)}$, $\mu^{(2)}$) is calculated based on demand volumes. Fig. 7 shows a graphical representation of the setups/cleanings. In addition, cleanings of forms are carried out with uncapacitated separable cleanings, and are not shown in the figure.

The precedence relationships among products are described next. Twelve products have due dates early in the production week: Monday to Thursday. This is because these are natural products, which are produced before products with stronger flavors, and because special treatment such as pricking or coating is required in further processing, which should not occur on weekends. Flavored cheeses such as those containing mushrooms or herbs are produced after the natural cheeses (δ_{fl}). Flavored cheeses need no special treatment during further processing, because the flavors have already been added, and therefore have the end of week as due dates. Strong flavored cheeses are produced after the flavored cheeses. These are six cheeses that must be produced at the end of the production week (δ_{fl}), because their recipes include garlic or hot peppers. After production of these strong flavored products an extensive cleaning is carried out. Solving the model for all products is computationally hard. We decompose therefore the problem based on the natural sequence of product groups and schedule each group sequentially. The makespan, changeover information, and form usage of previous groups is set as input for the subsequent group. Table 3 shows the product groups as well as the form types (χ_{lf}) and due dates (γ_i) of products.

The constraints implemented for the case study are summarized next. Constraints (1, 2, 4, 5) are implemented, in which the makespan is defined (as with the dairy) as the completion time on machine 1, which is analogous to the makespan on machine 2 due to the no-wait condition. Constraints (6) are not required because groups are solved separately. Batch-dependent changeovers and timing on machines are implemented with no-wait condition (7, 9, 12, 37–39). Time-window-based cleanings are modeled with constraints (15–20) and exact cleaning with constraints (B1, B2). There are no volume-dependent cleanings. The corresponding tightening constraints (31–34) and variable domains (35, 36) are also implemented. Furthermore, tracking form usage is implemented with constraints (40–51). We apply the model to a case at a medium-sized German cheese dairy. The model is implemented and solved in IBM ILOG CPLEX Optimization Studio 12.6 on a 2.6 GHz Intel Xeon CPU with 32 GB RAM.

5.4. Illustrative results and model performance

Around 3300 to 3800 variables and around 3200 to 6800 constraints result depending on the product group. Maximum runtime is set to 20 minutes for each group and the optimality gap is reported. The optimality gap is the difference between a theoretical upper bound on the optimal objective value and the achieved objective value. The model is first solved for a particular demand week (week 1) and illustrative results are presented. Group 1 is solved to 2.4% optimality gap, group 2 to 2.1% optimality gap, and

Table 3

Product groups, form types, and due dates.

Group ¹	Product number	Form type	Due date
1	1	1	Thursday
1	2	1	Thursday
1	3	1	Monday
1	4	1	Monday
1	5	1	Thursday
1	6	2	Thursday
1	7	2	Tuesday
1	8	2	Tuesday
1	9	2	Thursday
1	10	3	Wednesday
1	11	3	Monday
1	12	3	Thursday
2	13	1	–
2	14	1	–
2	15	1	–
2	16	1	–
2	17	2	–
2	18	2	–
2	19	2	–
2	20	2	–
2	21	3	–
2	22	3	–
2	23	3	–
2	24	3	–
3	25	1	–
3	26	1	–
3	27	1	–
3	28	2	–
3	29	3	–
3	30	3	–

¹) Group 1: Natural and blue cheeses.

Group 2: Flavored cheeses.

Group 3: Strong flavored cheeses.

group 3 to optimality in a few seconds. The resulting optimality gap is considered acceptable.

The schedule on machine 1 (coagulator) is shown in Fig. 8. A capacity utilization of 74–84% is achieved depending on the week-day. This is a very high utilization as it also includes the time-dependent cleanings (42 minutes and 4 hours) that are scheduled every day. Waiting times on machine 1 are only 4.5% of the makespan, which is attributable to the smart exploitation of processing-time differences and cleaning coordination. For example, as also illustrated in Fig. 3, a lot with short processing time is frequently followed by a lot with a longer processing time to minimize waiting. Also, time-dependent cleanings are scheduled when switching from a lot with long processing time to a lot with shorter processing time. The completion times of the product groups are also shown in Fig. 8, demonstrating the large production volume of group 1. Furthermore, the lot sizing of products is shown. Some products are produced in one lot, while others are in two or more lots due to setup and cleaning requirements as well as restriction on form capacities.

Fig. 9 illustrates the filling schedule on machine 2 (filler). Several lots with same form type are usually produced subsequently by virtue of longer sequence-dependent batch changeover when switching between form types. Coordination of cleanings is also

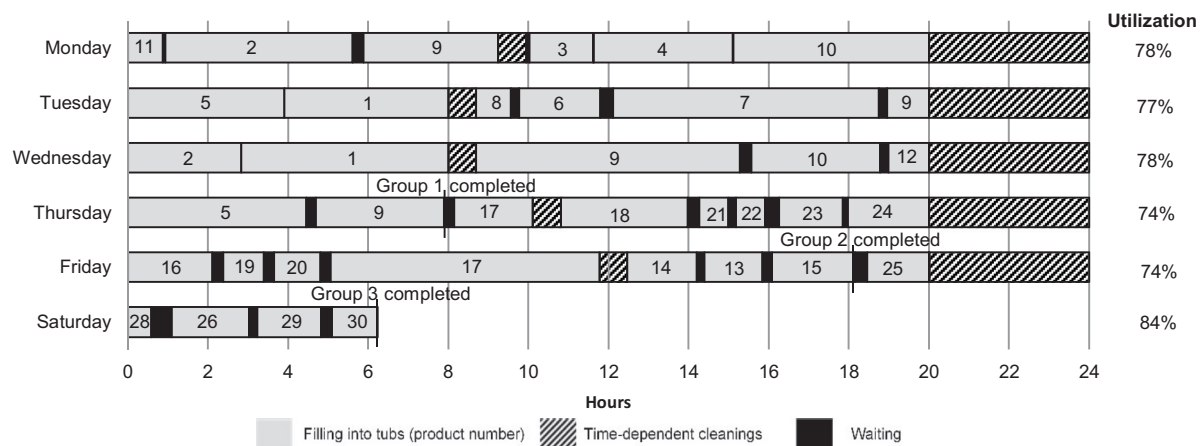


Fig. 8. Filling schedule on machine 1 (coagulator) for week 1.

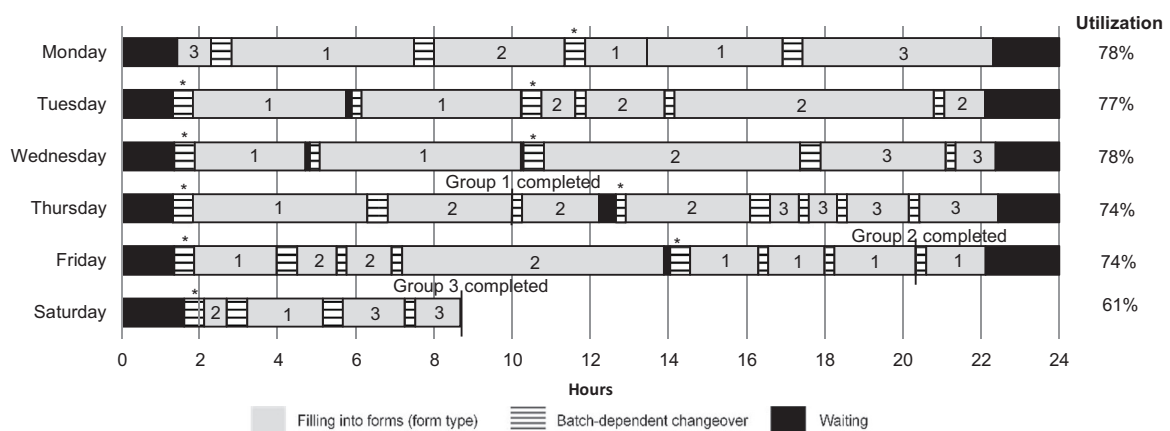


Fig. 9. Filling schedule on machine 2 (filler) for week 1.

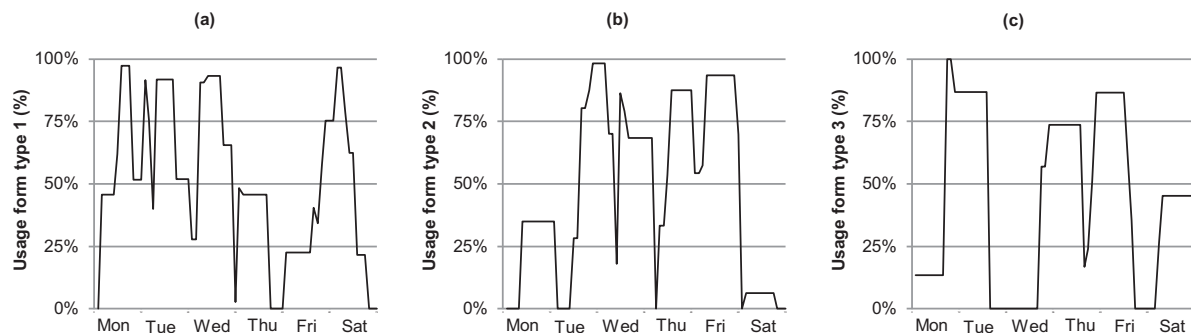


Fig. 10. Form usage for week 1; (a) form type 1, (b) form type 2, and (c) form type 3.

apparent in the results. Time-dependent cleanings are carried out concurrently with switching to a new form type in nine out of ten cases. In total, 35.8% of the batch-dependent changeover duration is carried out concurrently with time-dependent cleanings (marked with an asterisk in Fig. 9). Coordination of cleanings, therefore significantly reduces the waiting times of the no-wait flowshop and improves the performance of the model. Fig. 10 shows the form usage for each form type. The same form type is regularly filled until reaching the maximum form capacity after which filling of a new form type is started.

The performance of the developed approach is also compared to current planning at the cheese dairy (case study). The comparison is shown for machine 1, but machine 2 has the same utilization as machine 1 on all days except for Saturdays. Fig. 11, illustrates

the waiting times on machine 1 resulting from the developed approach and the planned schedule for the case study. For this demand week, a 50.9% reduction in waiting times is realized with the developed approach. Since the time-dependent cleanings are scheduled each day, they can only be reduced by finishing before their starting time. On the other hand, the waiting times on machine 1 can be reduced by smart exploitation of processing-time differences and by efficient scheduling of changeovers.

To validate the developed approach further, it is solved for two more demand weeks (weeks 2 and 3) and also compared to the planned schedule for the case study. The results of all weeks are summarized in Table 4, which shows the stable performance of the developed approach. On average, a 39.1% reduction in waiting times is achieved. Also, one short time-dependent cleaning is

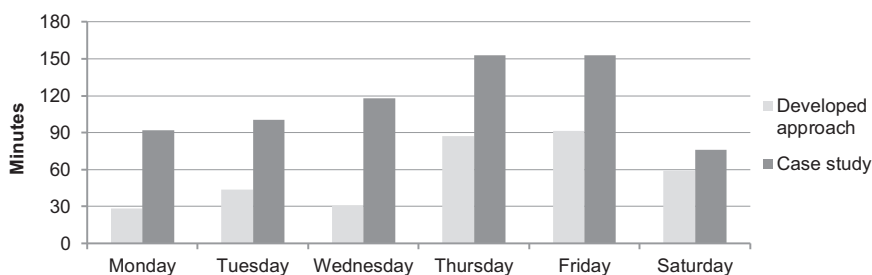


Fig. 11. Comparison of waiting times on machine 1 for demand week 1.

Table 4

Comparison of the developed approach and the planned schedule for the case study.

	Approach	Week 1	Week 2	Week 3
Optimality gap (%)	Developed approach	2.4; 2.1; 0.0	2.5; 3.0; 0.0	3.5; 1.8; 0.0
(Group 1; Group 2; Group 3)	Case study	–	–	–
Makespan on machine 1 (minutes)	Developed approach	7574.0	7864.5	7850.8
	Case study	7968.3	8146.2	8061.0
<i>Downtimes</i>				
Waiting on machine 1 (minutes)	Developed approach	340.0	370.5	450.4
	Case study	692.3	610.2	618.6
Time-dependent cleanings on machine 1 (minutes)	Developed approach	1410.0	1410.0	1410.0
	Case study	1452.0	1452.0	1452.0

saved in each week leading to a 2.9% reduction in time-dependent cleaning duration. This results in an average of 14.1% reduction in downtimes of machine 1 and a 3.7% makespan improvement. Significant cost reduction is thus realized by utilizing the developed approach, because the plant can be closed on average five hours earlier each week and less must be paid in overtime salaries on weekends.

5.5. Managerial insights based on sensitivity analyses

Three numerical tests are analyzed in this section. In test 1, the restriction on form capacities is relaxed. The aim is to gain managerial insights on how much the limited form capacities impact the schedule. This may provide insight in the need for investments. In test 2, time-window-based cleanings that are scheduled daily after 8–12 hours of production in the original settings must start daily at an exact time point (after 10 hours production). The aim is to analyze the value of providing the organizational flexibilities in scheduling cleanings and of employing a planning approach that is able to exploit it. In test 3, heterogeneous processing times on machine 1 are replaced with homogenous processing times for all products. Here, the aim is to analyze, if management should strive to streamline the product portfolio such that the processing times become similar. The homogenous processing times are calculated as weighted averages of the heterogeneous processing times (122.9 minutes for week 1, 122.8 minutes for week 2 and 122.2 minutes for week 3). Tests 2 and 3 are conducted with relaxed form capacities and compared to the first test, such that restricted form capacities do not interfere with the results. In all tests, the runtime is set to 20 minutes for each group, as it is in Section 5.4. Yet group 3 is always solved to optimality in few seconds. Table 5 summarizes the main results.

Comparing the results of test 1 with the original setting in which form capacities are restricted (see Table 4) shows an average makespan improvement of 0.8% and waiting time reduction of 15.6%. Fig. 12 illustrates the form usage of test 1 for week 1. The results show that for relaxed form capacities, more forms are frequently required than the maximum capacity, i.e., above 100% usage. This is the case for all three form types. For example, many lots requiring form type 3 are produced sequentially on Tuesday reaching a usage of 144%. Furthermore, for relaxed form capacities,

products are produced in fewer lots with larger lot sizes. The same pattern is realized for weeks 2 and 3. This shows that limited form capacities greatly impact the schedule. Therefore, the cheese dairy should analyze the possibilities of further investments.

The results of test 2, show that by replacing the time-window-based cleaning with an exact cleaning, an average waiting time increase of 10.2% is realized compared to test 1. Average downtimes increase by 2.7%, because with rigid starting times there is no flexibility when scheduling cleanings. Providing flexibilities in scheduling cleanings is therefore of great value, if the planning method is able to exploit it. The importance of accurately identifying the setup and cleaning classes is also apparent.

The results of test 3 show that homogenous processing times lead to a significant increase in makespan (1.7% on average) compared to heterogeneous processing times. Average waiting times increase by 41.9% and machine downtimes by 7.6%. This substantial increase in waiting times results, because smart exploitation of processing-time differences to schedule setups and cleanings is not possible. Fig. 13 illustrates this with excerpts from the schedules for week 1 for both heterogeneous and homogeneous processing times. During this week, the same products are produced in the first two lots. A form change between these lots is required on machine 2. The figure illustrates how the processing-time difference is exploited in the case of heterogeneous processing times to schedule the form change, which leads to less waiting time on machine 1. To summarize, a portfolio of products with heterogeneous processing times, as compared to homogenous processing times, is advantageous.

6. Conclusions

Extensive requirements for setups and cleanings exist in the process industries. These result in prolonged machine downtimes, waste of material, and high utility consumption in terms of energy, water, and cleaning agents. Efficient scheduling of setups and cleanings is therefore fundamental. This paper develops a classification scheme that distinguishes three different classes of setups and cleanings: batch-, time-, and volume-dependent setups and cleanings. Each class is further differentiated based on general characteristics such as separability, substitutability, reference point, and flexibility. In addition, setups and cleanings are

Table 5
Results of numerical tests.

	Test number ¹	Week 1	Week 2	Week 3
Optimality gap (%) (Group 1; Group 2; Group 3)	1	1.1; 2.1; 0.0	1.3; 2.2; 0.0	1.6; 2.0; 0.0
	2	1.7; 2.3; 0.0	1.7; 2.6; 0.0	2.5; 2.2; 0.0
	3	2.9; 2.7; 0.0	2.1; 2.5; 0.0	2.5; 2.7; 0.0
Makespan on machine 1 (minutes)	1	7518.2	7852.5	7728.3
	2	7550.5	7923.8	7765.0
	3	7686.4	7915.2	7889.2
<i>Downtimes</i>				
Waiting on machine 1 (minutes)	1	284.2	358.5	327.9
	2	316.5	387.8	364.6
	3	452.4	421.2	488.8
Time-dependent cleanings on machine 1 (minutes)	1	1410.0	1410.0	1410.0
	2	1410.0	1452.0	1410.0
	3	1410.0	1410.0	1410.0

¹) Test 1: Relaxed form capacities.

Test 2: Relaxed form capacities and exact scheduling of both time-dependent cleanings.

Test 3: Relaxed form capacities and homogeneous processing times.

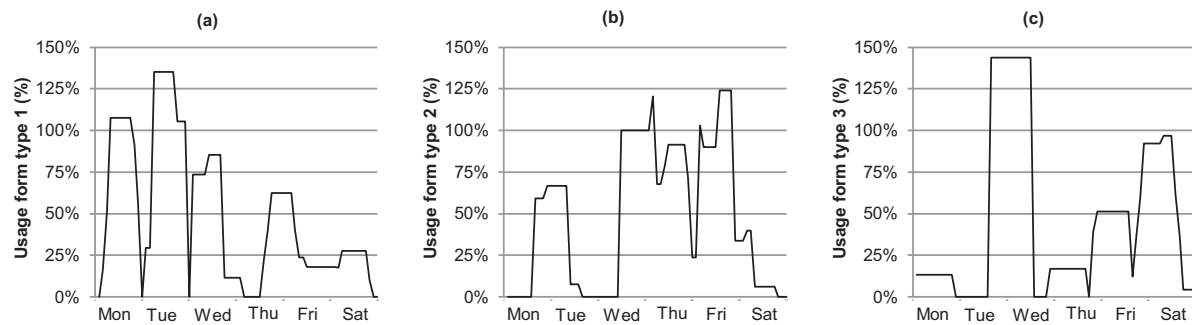


Fig. 12. Form usage for week 1 with relaxed form capacity restriction; (a) form type 1, (b) form type 2, and (c) form type 3.

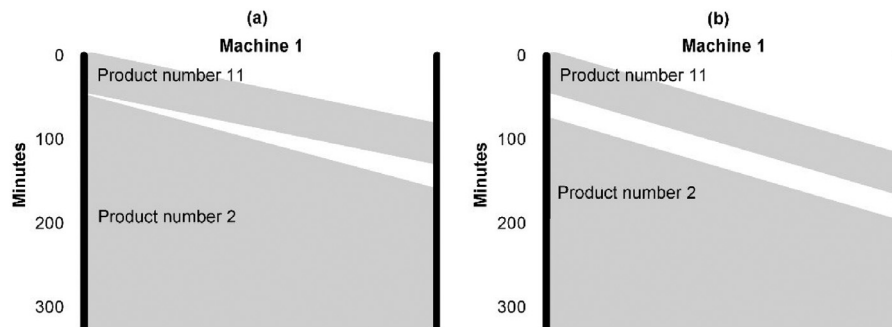


Fig. 13. Excerpts from schedules for week 1; (a) heterogeneous processing times and (b) homogeneous processing times.

characterized according to the determinants of the setup and cleaning time and/or costs, distinguishing product dependency from batch-size dependency.

The developed classification scheme facilitates the representation of setups and cleanings in the modeling of lot sizing and scheduling problems. Setups and cleanings are especially important in the process industries, in which flowshops are the prevalent type of production system. Therefore, a generic lot sizing and scheduling modeling approach for flowshops is developed based on the classification scheme. Besides precedence relationships and heterogeneous processing times, all setup and cleaning classes with various characteristics as well as potential substitution relationships between different setups and cleanings are accurately represented in the model.

The model is applied to a case study at a medium-sized German cheese dairy involving several classes of setups and cleanings. Sequence-dependent batch changeover and two different time-dependent cleanings must be scheduled. The cheese types have heterogeneous processing times and are connected through prece-

dence relationships. A no-wait condition must be employed between the machines, whereby lots are immediately loaded onto the next machine. In the cheese industry, a specific requirement concerns the use of different types of cheese forms. The forms are blocked for several hours after filling and represent a key bottleneck in the production of cheeses. The general model is adapted to account for the associated capacity considerations and the no-wait condition. These adaptations of the general model to the industry-specific requirements can easily be done, because the solution approach is based on a MILP formulation in which algebraic expressions can be altered and added while maintaining computational efficiency.

The numerical results of the case study demonstrate the applicability and the advantages of the developed approach. The cheese dairy realizes significant cost reductions by utilizing the developed approach. An average reduction in machine downtime of 14.1% compared to the outcome of their current manual lot sizing and scheduling process is achieved, which results in a 3.7% makespan improvement. Each week, the cheese dairy can close the plant on

average five hours earlier thereby significantly reducing overtime on weekends.

The numerical results show that the model is able to exploit the flexibility of the production system and even benefit from its complexity. However, accurate setup and cleaning class identification is prerequisite. More specifically, not allowing for flexible starting times of cleaning operations may ease planning and execution but over-constrains the solution space. Our results indicate that rigid starting times lead to a 2.7% increase in machine downtimes. Similarly, homogeneous processing times may be easier to handle in manual scheduling. However, the additional complexity that heterogeneity in processing times entails also goes along with the possibility to compensate for setup- and cleaning-time differences in the successive production stages. Our results show that neglecting heterogeneous processing times leads to a 7.6% increase in machine downtimes.

This paper developed a generic lot sizing and scheduling modeling approach for two-machine flowshops, which are common in large parts of the process industries. The model could easily be adapted such that more machines, and their specific setup and cleaning requirements, are included. Further research could also extend the numerical analyses to other industrial applications, because the developed approach provides adaptability to industry-specific features. While the developed models could be solved with sufficiently small optimality gaps using standard solvers, other industries may require the development of more elaborate solution methodologies. Analyzing the computational effort for different combinations of constraints in the developed modeling approach, could also be a subject for future research. Future research could also explore the use of the developed classification scheme in the development of solution approaches for other production environments. An example might be the make-and-pack systems that are frequently found in the food sector and also require different types of cleanings.

In addition to the economic impacts, setups and cleanings also have significant environmental impacts occasioned by material and utility consumption and the corresponding waste and emissions. In the case of the food industry, cleanings for instance contribute substantially to industrial food waste. Even though our approach efficiently utilizes setups and cleanings and thus implicitly reduces related environmental impacts, a more explicit consideration of all relevant environmental impacts may be desirable.

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References

- Allahverdi, A. (2016). A survey of scheduling problems with no-wait in process. *European Journal of Operational Research*, 255(3), 665–686.
- Allahverdi, A., Gupta, J. N. D., & Aldowaisan, T. (1999). A review of scheduling research involving setup considerations. *Omega*, 27(2), 219–239.
- Allahverdi, A., Ng, C. T., Cheng, T. C. E., & Kovalyov, M. Y. (2008). A survey of scheduling problems with setup times or costs. *European Journal of Operational Research*, 187(3), 985–1032.
- Allahverdi, A., & Soroush, H. M. (2008). The significance of reducing setup times/setup costs. *European Journal of Operational Research*, 187(3), 978–984.
- Amorim, P., Pinto-Varela, T., Almada-Lobo, B., & Barbósa-Póvoa, A. P. F. D. (2013). Comparing models for lot-sizing and scheduling of single-stage continuous processes: operations research and process systems engineering approaches. *Computers & Chemical Engineering*, 52, 177–192.
- Berlin, J., Sonesson, U., & Tillman, A.-M. (2007). A life cycle based method to minimise environmental impact of dairy production through product sequencing. *Journal of Cleaner Production*, 15(4), 347–356.
- Buschkuhl, L., Sahling, F., Helber, S., & Tempelmeier, H. (2010). Dynamic capacitated lot-sizing problems: a classification and review of solution approaches. *OR Spectrum*, 32(2), 231–261.
- Bylund, G. (2003). *Dairy processing handbook*. Lund: Tetra Pak Processing Systems AB.
- Cheng, T. C. E., Gupta, J. N. D., & Wang, G. (2000). A review of flowshop scheduling research with setup times. *Production and Operations Management*, 9(3), 262–282.
- Claassen, G. D. H., Gerdessen, J. C., Hendrix, E. M. T., & van der Vorst, J. G. A. J. (2016). On production planning and scheduling in food processing industry: Modelling non-triangular setups and product decay. *Computers & Operations Research*, 76, 147–154.
- Claassen, G. D. H., & van Beek, P. (1993). Planning and scheduling packaging lines in food industry. *European Journal of Operational Research*, 70(2), 150–158.
- Clark, A. R., Morabito, R., & Toso, E. A. V. (2010). Production setup-sequencing and lot-sizing at an animal nutrition plant through ATSP subtour elimination and patching. *Journal of Scheduling*, 13(2), 111–121.
- Copil, K., Wörbelauer, M., Meyr, H., & Tempelmeier, H. (2017). Simultaneous lotsizing and scheduling problems. *A Classification and Review of Models. OR Spectrum*, 39(1), 1–64.
- Doganis, P., & Sarimveis, H. (2007). Optimal scheduling in a yogurt production line based on mixed integer linear programming. *Journal of Food Engineering*, 80(2), 445–453.
- Doganis, P., & Sarimveis, H. (2008). Optimal production scheduling for the dairy industry. *Annals of Operations Research*, 159(1), 315–331.
- Floudas, C. A., & Lin, X. (2004). Continuous-time versus discrete-time approaches for scheduling of chemical processes: a review. *Computers & Chemical Engineering*, 28(11), 2109–2129.
- Gellert, T., Höhn, W., & Möhring, R. H. (2011). Sequencing and scheduling for filling lines in dairy production. *Optimization Letters*, 5(3), 491–504.
- Günther, H.-O. (2014). The block planning approach for continuous time-based dynamic lot sizing and scheduling. *Business Research*, 7(1), 51–76.
- Günther, H.-O., van Beek, P., Grunow, M., Lütke Entrup, M., & Zhang, S. (2006). An MILP modelling approach for shelf life integrated planning and scheduling in scaled sausage production. In M. Morlock, C. Schwindt, N. Trautmann, & J. Zimmermann (Eds.), *Perspectives on operations research* (pp. 163–188). Wiesbaden: DUV.
- Gupta, J. N. D., & Stafford, E. F. (2006). Flowshop scheduling research after five decades. *European Journal of Operational Research*, 169(3), 699–711.
- Hall, N. G., Laporte, G., Selvarajah, E., & Sriskandarajah, C. (2003). Scheduling and lot streaming in flowshops with no-wait in process. *Journal of Scheduling*, 6(4), 339–354.
- Hall, N. G., & Sriskandarajah, C. (1996). A survey of machine scheduling problems with blocking and no-wait in process. *Operations Research*, 44(3), 510–525.
- Hejazi, S. R., & Saghaian, S. (2005). Flowshop-scheduling problems with makespan criterion: a review. *International Journal of Production Research*, 43(14), 2895–2929.
- Kallrath, J. (2002). Planning and scheduling in the process industry. *OR Spectrum*, 24(3), 219–250.
- Kilic, O. A. (2011). Scheduling a two-stage evaporated milk production process. In O. A. Kilic (Ed.), *Planning and scheduling in process industries considering industry-specific characteristics* (pp. 67–95). Groningen: University of Groningen. Ph.D. Thesis.
- Kim, K., & Jeong, I.-J. (2009). Flow shop scheduling with no-wait flexible lot streaming using an adaptive genetic algorithm. *The International Journal of Advanced Manufacturing Technology*, 44(11), 1181–1190.
- Kopanos, G. M., Puigjaner, L., & Georgiadis, M. C. (2010). Optimal production scheduling and lot-sizing in dairy plants: the yogurt production line. *Industrial & Engineering Chemistry Research*, 49(2), 701–718.
- Kopanos, G. M., Puigjaner, L., & Georgiadis, M. C. (2011). Resource-constrained production planning in semicontinuous food industries. *Computers & Chemical Engineering*, 35(12), 2929–2944.
- Kopanos, G. M., Puigjaner, L., & Georgiadis, M. C. (2012). Efficient mathematical frameworks for detailed production scheduling in food processing industries. *Computers & Chemical Engineering*, 42, 206–216.
- Kumar, S., Bagchi, T. P., & Sriskandarajah, C. (2000). Lot streaming and scheduling heuristics for m-machine no-wait flowshops. *Computers & Industrial Engineering*, 38(1), 149–172.
- Lin, B. M. T., & Cheng, T. C. E. (2001). Batch scheduling in the no-wait two-machine flowshop to minimize the makespan. *Computers & Operations Research*, 28(7), 613–624.
- Lütke Entrup, M., Günther, H.-O., van Beek, P., Grunow, M., & Seiler, T. (2005). Mixed-integer linear programming approaches to shelf-life-integrated planning and scheduling in yoghurt production. *International Journal of Production Research*, 43(23), 5071–5100.
- Méndez, C. A., Cerdá, J., Grossmann, I. E., Harjunkoski, I., & Fahl, M. (2006). State-of-the-art review of optimization methods for short-term scheduling of batch processes. *Computers & Chemical Engineering*, 30(6–7), 913–946.
- Menezes, A. A., Clark, A., & Almada-Lobo, B. (2011). Capacitated lot-sizing and scheduling with sequence-dependent, period-overlapping and non-triangular setups. *Journal of Scheduling*, 14(2), 209–219.
- Shingo, S. (1985). A revolution in manufacturing. *The SMED system*. Portland, Oregon: Productivity Press.
- Sriskandarajah, C., & Wagneur, E. (1999). Lot streaming and scheduling multiple products in two-machine no-wait flowshops. *IIE Transactions*, 31(8), 695–707.
- Tempelmeier, H., & Buschkuhl, L. (2008). Dynamic multi-machine lotsizing and sequencing with simultaneous scheduling of a common setup resource. *International Journal of Production Economics*, 113(1), 401–412.

- Tempelmeier, H., & Copil, K. (2015). Capacitated lot sizing with parallel machines, sequence-dependent setups, and a common setup operator. *OR Spectrum*, 38(4), 819–847.
- Van Elzakker, M. A. H., Zondervan, E., Raikar, N. B., Grossmann, I. E., & Bongers, P. M. M. (2012). Scheduling in the FMCG industry. An industrial case study. *Industrial & Engineering Chemistry Research*, 51(22), 7800–7815.
- Walstra, P., Wouters, J. T. M., & Geurts, T. J. (2010). *Dairy science and technology* (2nd ed.). Boca Raton: CRC press.
- Wang, X., & Cheng, T. C. E. (2006). A heuristic approach for two-machine no-wait flowshop scheduling with due dates and class setups. *Computers & Operations Research*, 33(5), 1326–1344.
- Yang, W.-H. (1999). Survey of scheduling research involving setup times. *International Journal of Systems Science*, 30(2), 143–155.